SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: [Insert date]

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick “No” or write “None” – do not leave it blank. Submit as a single PDF document named “SDC2026_TeamName_TechDeclaration.pdf” together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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1. TEAM AND PROJECT OVERVIEW

Team Name

Chilli Pan MEE Penang

Project / Solution Name

Liggy

Name (Leader)

Tai Kai Xuan

Email & Phone (Leader)

elisebelletai1228@gmail.com

University Affiliation(s)

Universiti Teknologi Malaysia / Taylor's University

Live Prototype URL

https://sdchackathonproject-2.replit.app

Code Repository URL (public / access granted to judges)

https://github.com/CHUNG-WEI-YONG/SDC_HACKATHON_PROJECT

2. ARCHITECTURE & TECHNOLOGY STACK

2.1. System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Liggy is a small-business e-invoicing application consisting of a React frontend, a Node.js and Express backend, and a Python-based invoice agent. The frontend communicates with the Express API, which launches the Python agent to process invoice information using a configurable OpenAI-compatible AI service.

Users can manually enter invoice information or upload PDF, image, TXT, JSON, or CSV documents. The AI agent extracts and processes invoice details, which users can review, correct, approve, or reject. Approved invoices are generated as PDF files and their records are stored in Google Cloud Firestore.

Firestore also stores user accounts and customer information. The application provides customer management, invoice history, daily sales reports, monthly insights, and an AI help chatbot. Optional SMTP configuration supports emailing approved invoice PDFs.

The README documents local development using Vite on port 5173 and Express on port 3001. A public deployment URL and the actual configured AI provider and model are not documented.

2. ARCHITECTURE & TECHNOLOGY STACK

2.2. Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
JavaScript / JSX; React ^19.1.0; React DOM ^19.1.0	Developing Liggy's interactive frontend user interface and invoice management dashboard.	React & React DOM: MIT License; JavaScript/JSX: N/A (programming language/syntax)
Node.js 20+; Express ^5.1.0; cors ^2.8.5; dotenv ^16.5.0; nodemon ^1.14.10	Backend API development, HTTP request handling, environment configuration, and local development.	Node.js, Express, cors & nodemon: MIT License; dotenv: BSD 2-Clause License
Vite ^6.3.5; @vitejs/plugin-react ^4.6.0	Frontend development, React integration, and application bundling.	MIT License (both)
firebase-admin ^14.5.0; openai ^5.8.2; multer ^2.4.0; pdf-parse ^1.1.1	Firestore integration, communication with OpenAI-compatible AI endpoints, and document-related dependencies.	firebase-admin & openai: Apache License 2.0; multer & pdf-parse 1.x: MIT License
Python 3.10+; LangGraph >=0.2.0; langchain-core >=0.3.0; langchain-openai >=0.2.0; Pydantic >=2.0.0	Multi-agent workflow orchestration, GLM-5.3 integration through RunPod, and structured invoice data processing.	Python: PSF License; langgraph, langchain-core, langchain-openai & pydantic: MIT License
pypdf >=4.0.0; Pillow >=10.0.0; python-dotenv >=1.0.0; firebase-admin >=6.5.0; ReportLab >=4.0.0	Receipt and PDF processing, image handling, configuration, database integration, and standardized invoice PDF generation.	pypdf & python-dotenv: BSD 3-Clause License; Pillow: HPND / MIT-CMU (version-dependent); firebase-admin: Apache License 2.0; reportlab: BSD License

3. APIs & CLOUD SERVICES

3.1. External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API and Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
RunPod Cloud API Service — RunPod	Provides GLM-5.3 model inference through OpenAI-compatible endpoints to support receipt interpretation, structured invoice extraction, AI-assisted processing, and interactive chatbot responses.	Sponsor-provided

3. APIs & CLOUD SERVICES

3.2. Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit — Replit	Frontend interface development, application prototyping, and live web application hosting.	Core Plan
RunPod Cloud Platform — RunPod	Hosts the GLM-5.3 language model and provides cloud-based AI inference for Liggy's multi-agent invoice processing workflow and chatbot.	Sponsor-provided
Google Cloud Firestore — Google / Firebase	Cloud database for user accounts, customer information, approved invoice records, and transaction history.	Free Tier
SMTP Email Service — User-configured provider	Enables authorized email delivery of generated PDF e-invoices to customers using user-provided SMTP configurations	

4. AI TOOLS, MODELS & DATASETS

4.1. AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
GLM-5.3	GLM-5.3 model accessed through RunPod Cloud Platform and API Service	Pre-trained	GLM-5.3 License (Z.ai); subject to RunPod Terms of Service

4. AI TOOLS, MODELS & DATASETS

4.2. Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Customer and invoice transaction records	Google Cloud Firestore	Private application data		Y
User-uploaded payment receipts and invoice documents	Documents uploaded directly by application users	Private / user-provided		Y
User-authorized email and SMTP configuration data	Configuration information supplied by users	Private / user-provided		Y

4. AI TOOLS, MODELS & DATASETS

4.3. AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
ChatGPT	Implementation of code	Moderate

4. AI TOOLS, MODELS & DATASETS

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

To ensure accuracy and accountability for AI-generated outputs, our team implemented a two-layer verification workflow combining multi-agent cross-checking with human-in-the-loop oversight. First, after our initial extraction agent analyzes an uploaded receipt image or PDF to gather billing data, a separate, independent validation agent acts as a third-party auditor to review the parsed fields, verify mathematical totals, and flag potential extraction errors. Second, before any official e-invoice is generated or saved to the database, a human manager must manually review, edit if necessary, and explicitly approve the validated information, ensuring full human responsibility over all finalized financial records.

5. SPONSOR-PROVIDED RESOURCES

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
RunPod	RunPod Cloud Platform and API Service	Provides the essential cloud infrastructure for hosting and accessing the GLM-5.3 model. The service supports receipt image and PDF interpretation, automated invoice information extraction, AI-assisted verification, and interactive chatbot communication through OpenAI-compatible API endpoints. It serves as the primary AI inference provider for Liggy's LangGraph multi-agent e-invoice generation system.

5. SPONSOR-PROVIDED RESOURCES

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

The sponsor-provided RunPod Cloud Platform and API Service is essential to our prototype's core functionality. Our automated e-invoice generation system relies entirely on AI inference hosted on RunPod to power every node in our multi-agent workflow. Specifically, the platform serves as the central engine that automatically analyzes uploaded receipt images or PDFs, cross-checks and reviews extracted payment data for accuracy and formats the validated information into the final standardized e-invoice layout. Without RunPod's high-speed API infrastructure powering our underlying GLM-5.3 agents, the end-to-end automated extraction, verification, and generation pipeline could not function.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.1. Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Our application collects user-uploaded payment receipts or scripts, customer email addresses, and user-authorized SMTP credentials to facilitate automated e-invoice generation and delivery. During processing, receipt data is transmitted via encrypted HTTPS requests to the RunPod API (GLM-5.3) for temporary extraction and validation. Once confirmed by the user, the finalized e-invoice record is stored in our local database and saved to a user-specified file path. Additionally, with explicit user permission, the system securely stores the user's email and SMTP service configuration to consistently automate sending generated PDF e-invoices directly to customers. All sensitive credentials and billing records are protected through access controls and encrypted communication channels. Receipt parsing data is held only temporarily in memory during active sessions, while finalized invoice records and authorized SMTP configurations are retained persistently for continuous service until the user revokes permission or deletes the data.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.2. Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Our system faces two primary technical limitations: provider dependency and visual hallucination. First, our multi-agent architecture relies heavily on the active RunPod AI server endpoint. If the remote server experiences downtime or instance termination, the system requires manually reconfiguring the base URL, model identifier, and API key to redeploy and restore service. We mitigate this by using the standardized OpenAI SDK wrapper, allowing rapid switching between compatible endpoints via environment configuration. Second, when processing low-resolution, blurry, or poorly lit receipt images, the vision-language model may hallucinate or misinterpret text and numerical figures, resulting in inaccurate billing data. To mitigate this risk, we implemented a two-tier verification workflow within LangGraph: an automated validation agent that cross-checks mathematical totals and flags anomalies, followed by a mandatory human-in-the-loop (HITL) step requiring explicit user confirmation before any PDF e-invoice is generated or saved.

7. COMPLIANCE & READINESS

Tick all boxes that apply before submitting.

- [/] All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- [/] All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- [/] No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- [/] Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- [/] Any use of official SDC sponsor resources has been declared in Section 5.
- [/] The prototype described here is accessible to judges through the links provided in Section 1.
- [/] All fields in this template have been completed within the word limits stated.

8. DECLARATION

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]



[Date]

24 October 2026